

S P E C T R A V I D E O
U S E R S G R O U P -- W g t n

Newsletter Number 8 -- March 1985

The next meeting is on Monday ,March 18th at 7.30 in the Staff Common Room , Wellington Clinical School . Mein Street Newtown.

The following meeting is on April 22 , same place and time.

Wgtn SV Users Group

Secretary / Newsletter Editor

Don Stanley Ph 896379 (hm) , 746906 (work)

Treasurer

Colin Chin Ph 325656 (hm)

Committee members

Garry Clark 367872

Ross Wakelin

Mike Dalliesi 848416

Dennis Timmons 837200

Darryl Alison 85522 (Paraparaumu) (hm) 399510 (wk)

Specific areas of responsibility for committee members are ...
Don -- secretary & newsletter ; Colin -- Treasurer & MSX developments ; Garry -- Radio Access & public domain tape software ;
Ross -- National Committee ; Mike -- overseas users groups ;
Dennis -- library ; Darryl -- program evaluation committee chairman & Kapiti users rep

New members . . .

Welcome to ... Tim Lidgett , Justin Kerr , Herb Grant , Mason and Jill Smith , Bob Curwood , G and A Smith , Iain Boyd , Rod Fergusson , Martin Lee, Tony Conroy, Steve Sinclair, Ross Fletcher , George Grant, B M Heyworth

Could any members not having received a membership card yet please let me know .

Number of Members = 120

Newsletter Contributions

If you are submitting programs to go in the newsletter they must satisfy the following conditions or they will NOT be printed.

- 1 ... be typed/printed on A4 paper with about 5 lines left blank at top and bottom of page
- 2 ... typed with a dark ribbon OR printed using emphasized print mode
- 3 ... include a description of what the program does , typed on A4 paper with example input
- 4 ... In the case of programs above 20 lines in length they should be sent on cassette , in this case there is no need for the printout as I can merge them into the newsletter. Enclose a stamped addressed envelope for return of cassette
- 5 ... line numbers should be continuous -- do a RENUM before printing the programs
- 6 ... PLEASE MAKE SURE THEY WORK . and would readers please note that I am NOT responsible for programs sent by people which do not work .
- 7 ... In the case of screen dumps from 3d plotters please indicate the function used and try to get the print as dark as possible -- not always easy i know .

All other newsletter contributions should be sent to me at P.O Box 7057 , Wellington South. They may miss being printed for a month if sent to any other box or given to anyone else. Articles should follow 1 and 2 above.

Deadline for Newsletter articles is the 30/31st of each month. Please do not send articles direct to the printer.

Discounts for members .. there seems to be some confusion about these , especially among members who do not attend meetings. All discounts offered by Moonshine Computers , Einstein Scientific and Microstyle Computers do NOT include hardware. Discounts may be offered on such peripherals as floppy disks, cassettes , books , furniture, printers , monitors but NOT on SpectraVideo hardware. Individual dealers may not offer all the goods above at discounted rates , inquire from the shops just what discounts are available to club members. Note that to obtain a discount on any goods you must produce your current membership card.

BASIC NOTES

More On MSX Basic vs SV Basic

The BASIC on your SV is an early version of the Microsoft standard, MSX. Here are the differences between the SV BASIC and the final MSX BASIC that I can find from the MSX manual.

MSX BASIC

There are 5 parameters to the screen command. They are (1) to set the screen mode, (2) to set the sprite size, (3) to turn the key click on or off, (4) to set the cassette baud rate and (5) to tell the system whether an MSX printer is attached.

There are 4 screen modes available.

There is a CALL statement which calls a routine from a ROM cartridge

Two variables VDP and BASE are in the specifications to allow access to the VDP registers and the pattern/sprite/color tables.

The commands PUT and GET to use with graphics are not present.

They are the only differences I can find from the MSX manual between the MSX minimim standard BASIC and the SV BASIC. There are sure to be extra commands in the different MSX machines, since the different manufacturers can incorporate drivers for peripherals they develop. Indeed there are a number of extra SV commands, all the MODEM commands for instance are there solely to drive SpectraVideos modem.

Also the PAINT command seems to have caused more than its fair share of controversy, with many people claiming that the SV command is different to the MSX one. WRONG. The MSX command calls for the ability to PAINT inside a border with a different colour to the border ONLY in MULTICOLOR mode, which is screen 2 on the SV. This is exactly as for the SV. The reason that PAINT will not do this in SCREEN 1 is due to the graphics chips inability to properly have two adjacent pixels in different colors under some circumstances.

Hardware and ROM wise there are several differences between

SV BASIC

There are 2 parameters to the screen command, equal to (1), (2) at left, and to switch the function key display on/off in text mode. SV Basic uses the more flexible CLICK ON/OFF Command to control the key click.

Only 3 screen modes can be accessed from BASIC.

No such facility exists.

These variables are not in the SV BASIC.

These commands are present.

MSX and SV. The ROM routines are in completely different places as are the I/O ports. Thus converting many MSX machine code programs would be very difficult. However my understanding of many of the cartridges is that they contain their own ROMS (as for SECTOR ALPHA) and would run on the SV with the cartridge adapter for MSX (MSX has a different sized slot than SV). Without either adapter or cartridge I can't test this out.

Hardware is more of a non-standard item than the software and most machines will have features different to the SV. However I have yet to see any MSX machine with a numeric keypad, which will make the SV 328 more attractive to many. Also the price of the SV 328 is \$895 at present, the same price as the Sony HIT_BIT MSX computer is rumoured to be costing here. This machine will have 2 MSX cartridge slots, which assumably means that a disk and extra memory could be attached but no other cartridge. It is a 64k machine and will have 3.5" drives (the MSX 'standard' calls for 3.5", 5.25" OR 8" drives which implies there can be NO disk standard under MSX unless all manufacturers choose the same size drives and same formats). There will be a builtin Centronics interface, but no numeric keypad. Evidently the cassette will be an additional purchase also. There is no mention of any 80 column capability in the brochures which will nullify any CP/M compatibility which the system may have if this is not available.

Also a further limitation of the MSX BASIC is that there is no facility for writing text on the graphics screens. SV seem to have provided this facility at the expense of the extra screen mode (a color text mode). (This information is from a review in the November issue of Your Computer)

GAMES SCORES

Spectron	9700	Alexander Lindsay	
Sector Alpha	666400	Gavin Knight	
Kung Fu	38170	Teane Clark	
	17696	Garry Clark (*)	
B C Force	748600	Deirdre Chin	
Telebunny	87810	Colin Chin	
	26470	Don Stanley (+)	
Omac	25700	Don Stanley (+)	
Ninja	64300 53600	Alexander Lindsay	NICOLA TUINMAN
Tetra	266600 256750	Colin Chin	NICOLA TUINMAN
Turboat	68390	Barth Thornton (*)	
Sasa	119240 74950	Alexander Lindsay	JOHN TUINMAN

(+) - started in normal level
 (*) - started in arcade level
 otherwise practice level or unknown

Extra Club Meetings - - - -

Kapiti owners ... a club meeting will be held in Paraparaumu on APRIL 13th (Saturday) from 1.00 to 4.00 . Please see the Kapiti Observer , Coastlands Notice Board and this newsletter for details of where .Several members will be on hand to demonstrate hardware ,software ,answer questions, and so on. → page 25

Hutt Valley owners ... likewise on MAY 11th same time at a venue to be advised ... Watch the newsletter for details .

So if you live out in the suburbs and can't usually attend meetings here's your chance --- tell your friends !!

Club Sponsors

Einstein Scientific -- 177 Willis Street , Wellington

MicroStyle Computers -- 198 Jackson Street , Petone

Back Copies of Magazine

Number 1 is completely gone , I will repeat a couple of articles from it in future issues. I have very few copies of numbers 2,3 and 4 left , but 5,6 and 7 are Ok at present. Back copies are \$1.00 each.

Program Evaluation Committee

What is it ... a 3 person committee who will evaluate your program and return comments on it to you

Why .. to provide software writers with unbiased comments on programs, particularly if you may wish to sell it later.

Who .. Darryl Alison , Andrew McMillen , Jonathan Clark

Where .. send programs on cassette PLUS DOCUMENTATION to Darryl at P.O Box 189 , Paraparaumu , or phone him with details.

and .. this committee will also provide a program assistance service. Ring Darryl if you have problems and he will put you on to the best person to try and solve it. This service will include problems with BASIC or CP/M and its components (eg DBASE2 , Wordstar , MBasic , 8080 Assembler , Pascal etc). If you want to be included as someone with particular skills who Darryl could

advise members to contact please give him your name , phone number and what you can help with .

When sending programs to Darryl please include details of what the program should do , whether it is disk/cassette/either , give a name for it , and whether you intend selling it commercially later , printing it in a magazine (Bits & Bytes etc) or whatever.

Question & Answer

Send your queries about anything SV to the editor at box 7057 and we will endeavour to answer it here ...

Question from Lester Reilly

When entering numeric input using the Basic INPUT statement how can I set the program up so that the user terminates data entry with just a return ?

Answer .. (ed)

Lester has a question here which BUGs everyone at some stage. The problem is that the INPUT statement filters out returns. Thus code like this in character input ..

```
10 input a$
20 if asc(a$) = 13 then end
30 goto 10
```

which allows just a return to halt the routine , has no obvious equivalent in numeric input. So you trick it a little bit. Choose an integer which can never be acceptable input , I use -9999 as I rarely need to use this as valid data. Now your input code becomes ...

```
10 a = -9999 : input a
20 if a = -9999 then end
30 goto 10
```

The trick here is to know that if a return is pressed with no other numeric input preceding then the last value given to a will still be a's value after the input , so if just a return is pressed a will be -9999 above and this is end of data.

Question from Kevin Wronski ...

If I use LPRINT TAB(n) .. in a program and finish the printed line with a CHR\$(10) (carriage return) then try to use a tab on the next line it does not work . Any suggestions ?

Answer .. (ed)

The answer is embodied in the question. CHR\$(10) is NOT a carriage return but a line feed. A line feed merely moves the printhead to the same position on the next line. A carriage return is CHR\$(13) but this leaves the printhead on the same line so you need to finish with a CHR\$(10) followed by a CHR\$(13) to get to the start of the next line, then the tab should work fine.

BUGS BUGS BUGS

The BUGS file is off to a flying start ... here are the ones received so far ...

- 1 ... BUG --- BASIC ROM GENERATES A SYNTAX ERROR WHEN A HEX NUMBER IS FOLLOWED BY A KEYWORD OR VARIABLE NAME BEGINNING WITH ANY OF THE CHARs A,B,C,D,E,F AND NO SPACE IS LEFT BETWEEN NUMBER AND WORD

E.G , POKE &HC000,&H12ELBE ... BOMBS

FIX -- ALWAYS LEAVE A SPACE BETWEEN HEX NUMBERS AND A KEYWORD LIKE ELBE

- 2 ... BUG --- BASIC GENERATES AN ILLEGAL FUNCTION CALL WHEN THE WORD AS IS FOLLOWED BY A CHARACTER VARIABLE NAME BEGINNING WITH C AND NO SPACE IS LEFT BETWEEN THEM

EB . FIELD#1,20ASC*

FIX ---AGAIN LEAVE A SPACE AFTER THE AS , THE STATEMENT IS TOKENISED INCORRECTLY OTHERWISE INTO THE ASC* FUNCTION

- 3 ... BUG --- IN CALCSTAR VERSION 1.45 /= IS USED FOR REPEATING CHARACTERS . THE CHARACTER FOLLOWING IS REPEATED ON THAT LINE . IF /= ONLY IS TYPED FOLLOWED IMMEDIATELY BY RETURN , THEN IT CRASHES AND DUMPS YOU BACK IN CP/M WITH THE LOSS OF YOUR WORKFILE.

FIX --- NO SOFTWARE FIX KNOWN , TAKE CARE NOT TO HIT RETURN UNDER THE ABOVE CIRCUMSTANCES.

New Hardware

No dates set yet for any of these , but there will be an SV networking system released this year , evidently able to accomodate up to 32 computers (318/328 or MSX) . A voice

synthesiser is in the pipeline , this is being developed by a New Zealand company and I gather is in a fairly advanced stage. Seen advertised in Australian Your Computer 1984 was a SpectraVideo cartridge and tape copier , I know no further details of this.

Also I spoke to Peter De Zwart from CDL recently , the MSX cartridge adapter is still somewhere in SpectraVideos R & D system .

Speaking of errors . . .

The last KIWIMBX magazine contained a number of errors , here are some corrections if anyone got confused .

A line number uses 2 bytes , not 4 . There are 2 more bytes used by the system to point to the memory location where the next line starts (page 4)

The first example on page 4 does use 6 bytes but the reasoning is wrong. The line number uses 2 bytes as above and 2 are used to point to the next line. There is a byte used by the character R but the 6th byte is not the space , this is inserted by the editor on the screen for readability, it uses no memory. The 6th byte is actually a null byte in the memory location which the 2 bytes pointing to next line indicated. A null byte here indicates end of program.

The PUT. SPRITE command on page 4 uses memory as follows .The line number and next line pointer use 4 bytes , the space between PUT & SPRITE uses 1 , PUT and SPRITE use 1 each and the 8th byte used is the null byte indicating end of program.

At the bottom of page 4 there is a limit on the number of commands on a single line , they may not exceed 255 tokenised characters

On page 3 it is stated that the computer abbreviates all command words to a single byte. This is wrong as many command words tokenise to 2 bytes.

On page 9 it is stated that PDL is not a SPECTRAVIDEO reserved word. It is , and controls a games paddle.

Program of the Month

This program was on a disk which arrived from Deane Whitmore. The object is to shoot (space bar , joystick trigger) the balls being fired at you. This program alters the VRAM contents while program is running to get some excellent graphics in here

```

10 COLOR15,0,0:SCREEN1,2:DEFINT A-Z:R=RDND(-TIME):CLEAR1000
20 F=1:G=0:C=50:A=4:B=1:GOSUB1010
30 F=49:G=48:C=80:A=4:B=4:GOSUB1010
40 F=79:G=78:C=90:A=4:B=5:GOSUB1010
50 F=89:G=88:C=98:A=5:B=5:GOSUB1010
60 F=77:G=96:C=107:A=7:B=4:GOSUB1010
70 F=106:G=105:C=110:A=7:B=7:GOSUB1010
80 F=109:G=108:C=110:A=8:B=9:GOSUB1010
90 F=110:G=111:C=192:A=13:B=5:GOSUB1010
100 PBET(0,98),1
110 DRAW"r4f1r2f1r1r4f3r1f6r1f2r3f1r8e1r6f1r2e2r3f1r1e2r2e3r2e
4r1e4r1e2r2e2r2e1r2f5r1f4r1f4r1f3r1f2r10e1r1e2r1e1r1e1r2e1
r2e2r1e1e1r1e1r1f2r2e1r1f3r1f1r2f3r1f2r2f1r4f2r3e3r1e3r1e3
r1e3r1e3r1e5r1e4r1e1r1e2r2e1r2e1r3f1r1f3r1f12r1f2r1f1r1f4r1f4r9e1"
120 LINE(0,120)-(256,120),1
130 PAINT(80,105),1
140 DRAW"bm173,110c11e3r1e3r1e3r1e5r1e5r1e4r1e1r1e2r2e1r2e1r3f
1d1g111g311g211g312g611g412g215g115":PAINT(192,100),11:
PAINT(200,93),11
150 DRAW"bm45,113r2e2r3f1r1e2r2e3r2e4r1e4r1e2r2e2r2e1r2f1d2g31
r2e12d1g213g212d1g313g114":PAINT(78,100),11:PAINT(58,111),11
160 DRAW"bm120,110e1r1e2r1e1r1e1r2e1r2e2r1e1e1r1e1r1f1g311g312g3
14":PAINT(128,106),11:PAINT(136,104),11
170 FOR Y=120 TO 190 STEP 17.5
180 LINE(127,120)-(0,Y),4
190 NEXT
200 FOR X=0 TO 256 STEP 20
210 LINE(127,120)-(X,190),4
220 NEXT
230 FOR Y=190 TO 120 STEP -23.2
240 LINE(127,120)-(256,Y),4
250 NEXT
320 Y=122:FOR X=0 TO 10
330 Y=X+Y:LINE(0,Y)-(256,Y),4,BF
340 NEXT
400 LINE(127,114)-(132,119),11,BF
1000 GOTO1060
1010 FOR Y=FTOCSTEP2
1020 LINE(0,Y)-(255,Y),A,BF:NEXT Y
1030 FOR Y=STOCSTEP2
1040 LINE(0,Y)-(255,Y),B,BF:NEXT Y
1050 RETURN
1060 FORT=1 TO 8:READ A:A=A+CHR$(A):NEXT:SPRITE$(0)=A$
1070 FORT=1 TO 32:READ B:B=B+CHR$(B):NEXT:SPRITE$(2)=B$
1080 GOSUB1400
1090 W=120:Z=180
1100 X=127
1110 D=INT(RND(1)*-6)+5:IF D=0 THEN 1110
1120 E=INT(RND(1)*-6)+5:IF E=0 THEN 1120
1130 F=INT(RND(1)*2):IF F=0 THEN Y=110 ELSE IF F=1 THEN Y=-110
1140 PUTSPRITE2,(W,Z),15,2
1150 B=STICK(0) OR STICK(1)
1160 K=STRIG(0) OR STRIG(1)
1170 SPRITEON:ONS PRITEGOSUB1450

```

```

1180 ONB080SUB1200,1210,1220,1230,1240,1250,1260,1270
1190 BOTO1280
1200 Z=Z-6:RETURN
1210 W=W+6:Z=Z-6:RETURN
1220 W=W+6:RETURN
1230 W=W+6:Z=Z+6:RETURN
1240 Z=Z+6:RETURN
1250 W=W-6:Z=Z+6:RETURN
1260 W=W-6:RETURN
1270 W=W-6:Z=Z-6:RETURN
1280 IFX<10RX>255THEN1390
1290 IFY>190ORY<10THEN1390
1300 IFY>125ORY<80THENB080SUB1370
1310 IFY>140ORY<30THENB080SUB1380
1320 X=X-D:Y=Y-E
1330 PUTSPRITE0,(X,Y),15,0
1340 BOTO1140
1350 DATA0,0,0,16,56,16,0,0
1360 DATA255,128,128,128,128,128,128,128,128,255,0,0,0,0,0,
255,1,1,1,1,1,1,1,1,255,0,0,0,0,0
1370 VPOKE14338,56:VPOKE14339,124:VPOKE14340,124:VPOKE14341,124:
VPOKE14342,56:X=X-D:D:Y=Y-E-E:RETURN
1380 VPOKE14337,124:VPOKE14338,254:VPOKE14339,254:VPOKE14340,254:
VPOKE14341,254:VPOKE14342,254:VPOKE14343,124:X=X-D-D:Y=Y-E-E:
RETURN
1390 VPOKE6912,209:VPOKE14337,0:VPOKE14338,0:VPOKE14339,16:
VPOKE14340,56:VPOKE14341,16:VPOKE14342,0:VPOKE14343,0:BOTO1100
1400 SR$=CHR$(0)+CHR$(18)+CHR$(9)+CHR$(105)+CHR$(23)+CHR$(94)+
CHR$(46)+CHR$(188)+STRING$(2,112)+CHR$(248)+CHR$(121)+CHR$(159)+
CHR$(41)+CHR$(36)+CHR$(0)+CHR$(8)+CHR$(80)+CHR$(105)+CHR$(224)+
CHR$(217)
1410 SPRITE$(3)=CHR$(2)+CHR$(30)+CHR$(30)+CHR$(59)+CHR$(240)+CHR$(
224)+CHR$(97)+CHR$(57)+CHR$(15)+CHR$(2)+STRING$(9,0)+STRING$(
4,128)
1420 SPRITE$(4)=8R$+CHR$(26)+CHR$(28)+CHR$(5)+STRING$(3,6)+CHR$(14)+
CHR$(13)+CHR$(217)+CHR$(118)+CHR$(140)
1430 SPRITE$(5)=CHR$(32)+CHR$(120)+CHR$(248)+CHR$(240)+CHR$(48)
1440 RETURN
1450 IFKTHEN1460ELSERETURN
1460 VPOKE6912,209:VPOKE6920,209:B080SUB1520:PUTSPRITE5,(X+1,Y-1),6,5:
FORT=1T080:NEXT:B080SUB1520:PUTSPRITE3,(X-2,Y-4),8,3
1470 FORT=1T050:NEXT:PUTSPRITE4,(X-6,Y-8),10,4:FORT=1T050:NEXT:
VPOKE6924,209:FORT=1T050:NEXT:PUTSPRITE3,(X-2,Y-4),8,3
1480 FORT=1T080:NEXT
1490 VPOKE6932,209:FORT=1T080:NEXT:VPOKE6924,209:FORT=1T050:NEXT:
VPOKE6928,209
1500 FORT=1T01000:NEXT
1510 SPRITEOFF:VPOKE6920,Z:BOTO1390
1520 SOUND 0,0:SOUND 6,30:SOUND 7,0:SOUND 8,16:SOUND 9,16:SOUND
10,16:SOUND 11,0:SOUND 12,5:SOUND 13,0:SOUND 12,26:RETURN

```

Public Domain Software

For those of you who requested public domain software from me and have not yet got it my apologies , I simply have not had time. We have revamped our public domain software system completely and it now runs as follows.

1) Do NOT send us tapes. We will supply a standard tape which will get round the problem of people sending tapes which are too short.

2) The cost of tapes will be \$3.00 each and we will also have a charge of \$2.50 to cover handling and return postage. This is charged for each separate lot of copying done , not per tape.

3) Any of the PD Software may also be broadcast on Radio Access in our 5 minute slot on the first Saturday of each month.

4) The tapes will contain combinations of cassette & disk programs , some may be set up for 80 columns. Adapt them as you wish to suit yourself.

5) Send orders with payment to PDS , P.O Box 7057 , Wgtn South, Wellington

6) Send us your public domain software for inclusion in the software library.

A full list of programs available will be printed next month .

A Note on ROM/RAM . . .

As you will be aware the BV328 memory is as follows

64k RAM , 16k Video Ram , 32k Rom , and if you own an 80 column card there is 4k Ram in as well . As you will also be aware when you boot up your system only 29k (cassette) or 21k (disk) is available to you.

The reason for this is the Z80 chip which can only access 64k at a time. It has to be able to reach the 32k ROM which leaves only 32k Ram accessible. A little more is lost due to the requirement for the system to have a work area in Ram which contains various flags and values used by ROM routines. BV memory is arranged in 32k pages , the Rom fills one page , leaving 32k available to be accessed which is filled by a page of Ram. This then leaves 32k Ram unable to be accessed from Basic.

Machine code programmers can access this page and use it for string storage and so on. Also if you want some extra memory and have a little time to sort out locations you can use the Video Ram for string storage (in text mode only of course) but the

usual means of accessing strings will not be available.

Serial Numbers

The possibility of burglary is always present , if members want the club to keep a list of serial numbers of hardware please fill out the last page form (NO NAMES , numbers only) and post it to SN , P.O Box 7057 , Wgtn South , Wellington.

Any members who purchased club manuals please send any corrections you have discovered to us ASAP as an update will be sent out shortly.

Also we are investigating the idea of rewriting the manual and selling it commercially. However the manual would form only a small part of such a venture , and we are seeking people who would be interested in contributing a chapter on ANY aspect of the system . We are looking to get about 10 -15 chapters so if you fancy your literary talents how about writing down what you would like to write about and sending it to me .

Heres a quick machine code program to set the function keys up for different lots of software in CP/M . (8080 code)

org 0100h

```

;
; DBASE 2 FKEYS definition
;
;key1 = set intensity
;key2 = set default to
;key3 = set talk off + crlf
;key4 = do menu+crlf
;key5 = quit
;key6 = display files on
;key7 = modify command
;key8 = set confirm on
;key9 = use svclub+crlf
;key10 = clear+crlf
;
avi      b,0          ; load key count at zero
ixi      d,key1       ; start of fk string in D
ixi      h,0fa1eh     ; fk locations in H
dumpdf:  xchg          ; swap D and H so can
mov      a,m          ; read from memory into a
xchg     a,h           ; and then swap them back
mov      m,a          ; move from a to memory
inx      h             ; up memory by one
inx      d             ; up FK string by one
inr      b             ; up counter by one
```

```

mov     a,b                ; move counter into A reg
cpl     0a0h              ; so can see if finished
rz      ; finished -- back to CPM
jmp     dumpdf             ; not finished -- continue
;
key1:   db 73h,65h,74h,20h,69h,6eh,74h,65h,6eh,73h,
          69h,74h,79h,20h,00h,00h
key2:   db 73h,65h,74h,20h,64h,65h,66h,61h,75h,6ch,
          74h,20h,74h,6fh,20h,00h
key3:   db 73h,65h,74h,20h,74h,61h,6ch,6bh,0dh,0ah,
          00h,00h,00h,00h,00h,00h
key4:   db 64h,6fh,20h,6dh,65h,6eh,75h,0ah,0dh,00h,
          00h,00h,00h,00h,00h,00h
key5:   db 71h,75h,69h,74h,00h,00h,00h,00h,00h,00h,
          00h,00h,00h,00h,00h,00h
key6:   db 61h,70h,70h,65h,6eh,64h,0ah,0dh,00h,00h,
          00h,00h,00h,00h,00h,00h
key7:   db 6dh,6fh,64h,69h,66h,79h,20h,63h,6fh,6dh,
          6dh,61h,6eh,64h,20h,00h
key8:   db 73h,65h,74h,20h,63h,6fh,6eh,66h,69h,72h,
          6dh,20h,6fh,6eh,20h,00h
key9:   db 75h,73h,65h,20h,73h,76h,63h,6ch,75h,62h,
          0ah,0dh,00h,00h,00h,00h
key10:  db 63h,6ch,65h,61h,72h,0ah,0dh,00h,00h,00h,
          00h,00h,00h,00h,00h,00h
;
;
end

```

To set the keys up to suit your own software (eg WORDSTAR) you use a text editor (ED, WORDSTAR, PEACHTEXT ...) to change the data lines which begin with the key number above. Just type the ASCII code you desire into the data lines (eg to make key 1 do a ^QY in WORDSTAR you would change key 1 data to read 0Fh,59h,00h and the other 13 bytes also become 00h). Save your changed program (eg fksetws.asm for WORDSTAR, fksetdb2.asm for DBASE2, fksetcpm.asm for general CPM keys) then use the CPM assembler ASM (ASM fksetws for example) to assemble the code. Assuming there were no errors you now use the CPM utility LOAD (LOAD fksetws for example), and then just type the name of the program (fksetws) to change the keys. They remain changed until you use a different program or reboot CPM.

Note that a null character terminates a function key, so you must have a 00h in each key definition above.

```

10 SCREEN,0
20 WIDTH40
30 LF$=CHR$(10)+CHR$(13)
40 RV$=CHR$(27)+"p":NV$=CHR$(27)+"q"
50 DIM A(15)
60 FOR I=1 TO 15
70 A(I) = 5000*RND(2)-2500*RND(1)
80 NEXT
90 CLS
100 GOSUB540
110 A$="#####.##":GOSUB650:GOSUB600
120 A$="+#####.##":GOSUB650:GOSUB 600
130 A$="#####.##-":GOSUB650:GOSUB 600
140 A$="*#####.##":GOSUB650:GOSUB 600
150 A$="#####.##":GOSUB650:GOSUB 600
160 A$="#####.##-":GOSUB650:GOSUB 600
170 A$="#####.##-":GOSUB650:GOSUB 600
180 A$="*#####.##-":GOSUB650:GOSUB 600
190 A$="#####.##-":GOSUB650:GOSUB 600
200 A$="#####.##^":GOSUB650:GOSUB 600
210 A$="#####.##^":GOSUB650:GOSUB 600
220 A$="#####.##^":GOSUB650:GOSUB 600
230 A$="#####.##^":GOSUB650:GOSUB 600
240 CLS
250 LOCATE7,0:PRINTRV$ Alpha Formats "NV$
260 PRINT:PRINTRV$ No Formats ... "NV$
270 A$="Example String "
280 LOCATE1,6:PRINT A$
290 LOCATE1,7:PRINT A$
300 LOCATE1,15:PRINT A$
310 LOCATE1,16:PRINT A$
320 LOCATE17,2:PRINT RV$ Format "NV$
330 LOCATE20,6:PRINT"!":LOCATE30,6:PRINT USING "!";A$
340 LOCATE20,7:PRINT"! ! !":LOCATE30,7:PRINT USING "! ! !";A$
350 LOCATE20,16:PRINT"\ ":LOCATE30,16:PRINT USING "\ ";A$
360 LOCATE20,15:PRINT"\ \ ":LOCATE30,15:PRINT USING "\ \ ";A$
370 LOCATE 0,20:PRINT " Example String was ..."
380 LOCATE 0,21:PRINT A$;
390 GOSUB570
400 CLS

```

CONT

DEND PROGRAM TO SHOW
PRINT USING with different formats

```

410 LOCATE5,0:PRINTRV$"      TAB and SFC      "NV$
420 LOCATE0,2:PRINT" TAB and SFC also help format output"
430 LOCATE0,4:PRINT"eg PRINT TAB(20)"CHR$(34)"Hello"CHR$(34);
440 PRINT"TAB(5)"CHR$(34)"AA"CHR$(34)
450 LOCATE0,6:PRINTRV$" Produces "NV$
460 PRINT:PRINT TAB(20)"hello"TAB(5)"AA"
470 LOCATE0,10:PRINT"eg PRINT SFC(20)"CHR$(34)"Hello"CHR$(34);
480 PRINT"SFC(5)"CHR$(34)"AA"CHR$(34)
490 LOCATE0,12:PRINTRV$" Produces "NV$:PRINT:PRINT SFC(20)"hello"SFC(5)"AA"
500 LOCATE 0,16:PRINTRV$" Note that TAB & SFC & USING can"
510 LOCATE 0,18:PRINT" only be used with the PRINT com-"
520 LOCATE 0,20:PRINT" mand"NV$;
530 END
540 LOCATE7,0:PRINT RV$" NUMERIC FORMATS "NV$
550 PRINT:PRINT RV$" No Formats ... "NV$
560 FOR I=1 TO 15:LOCATE0,1+4:PRINT A(I):NEXT
570 LOCATE5,24:PRINTRV$" Press Space Bar to Continue "NV$;
580 IF STRIG(0)=0 THEN 570
590 RETURN
600 FOR I=1 TO 15:LOCATE25,1+4:PRINT STRING$(16,32)
610 LOCATE25,1+4:PRINT USING A$;A(I):NEXT
620 GOSUB570
630 RETURN
640 WIDTH80
650 LOCATE19,2:PRINT RV$" Format "NV$:LOCATE27,2:PRINTA$;NV$:RETURN

```

```

*****
* EDUCATIONAL *
* SOFTWARE SOFTWARE *
* COMPETITION *
*****

```

In March the Club is running a software competition in conjunction with Radio Station ZYB.

What you have to do is to write an EDUCATIONAL program of less than 90 seconds loading time. Much smaller will be fine.

The closing date for entries is the 30th of April so this will give you enough time to perfect a program and come up with possibly original ideas.

All club members and radio listeners can participate and the judging will be done by Da... sorry thats a secret. (No one from the club). Prizes well thats a secret too except to say that Don & Barry will have to dig deep into their pockets if the committee disagrees with the proposed 1st and 2nd prizes. Sounds like an adventure game in the making!!

You may submit more than one program but it will be only possible for an individual to win one of the two prizes. It is hoped that all the best programs will be put on a cassette. Those whose programs make it to cassette will naturally receive free copies.

Send you Programs on cassette to: Access Radio ,Spectravideo Computer Programs,Box 2396,Wellington.

Please include your name and address on the cassette so that they can be returned.

Lets show them at Access Radio that SPECTRAVIDEO owners actually write programs on their computers. If this is a success there will be more competitions on different themes.

SPECTRAVIDEO USERS GROUP (Wgtn)

Membership Form

Name _____

Address _____

Phone (home) _____ (Work) _____

Interests (Please tick relevant ones)

Basic __ Assembler __ Games __ CP/M __ Education __

Work Use __ Other (please specify) __

Meetings (please tick relevant one)

Usually __ Sometimes __ Rarely __ Newsletter Only __

O F F I C E U S E O N L Y

Paid Date Expires Signed Number

Send Completed Form (Whole Page) plus \$10.00 to

SpectraVideo Users Group (WGTN)

c/- Don Stanley , P.O Box 7057 , Wellington South , Wellington

Many computer owners are successfully recording and running the computer programs. However there are a number who are experiencing problems.

The main causes of program loading failure are likely to be:

- 1) Poor quality cassette tape or trying to record data onto leader tape.
- 2) The signal strength of the radio too low or signal masked by electrical interference.
- 3) Alignment or dirty heads on cassette recorders.
- 4) Using Dolby or other noise reduction systems to process the sound onto and off the tape.

Of the above list, two require further comment.

SIGNAL STRENGTH: For the club members who live in Masterton, Palmerston Nth or New Plymouth you will be on the fringe of 2YB's coverage. Hilly terrain or the presence re-inforced or steel framed buildings and commercial areas can effect ones reception. To increase your signal strength try putting up an outside aerial of up to 6 metres.

AZIMUTH ALIONMENT of cassette record / replay head: This would be the most common reason why programs will not load into computers apart from tape drop-outs. Spectravideos problem with the radio broadcasts is that seprate cassette machines are used for recording and replaying data. If you are already successfully loading commercially produced tapes into your computer, then your SV903 or SV904 would appear to be set reasonably accurately. The cassette recorder you are using for recording the radio with will probably need a head alignment. You can set this recorder to the same azimuth as the SV data cassette by using a tape you have 'saved' yourself. PLAY the 'saved' tape on the radio cassette recorder and adjust the head azimuth with a small non magnetised screw driver. (Small screw located usually to the left of the record / replay head). Note the starting position before turning and adjust to produce the brightest and the maximum sound level possible. Normally you should not have to adjust the azimuth screw more than one turn, either way from the start. To gain access to the screw, on some machines the loading hatch cover will have to be removed or some times a small slot or hole near the head will indicate the screws position below. If you are recording on a stereo machine then it would be best to monitor in mono to make the adjustment more discernable.

If in doubt seek expert advice or bring your deck to the club meeting and I will set it for you. Remember bring a tape you have 'saved' yourself and also record a non data segment. This is a check that the head is not set to a false peak. To have the replay treble control set to maximum, will aid this method as you are tuning in recorded noise(hiss).

IMPROVEMENTS TO "SPECTRA FONT EDITOR"

by Donald Rogers

This 243-line program by Steven R Boal is for redesigning the characters which appear on the screen. It makes good use of graphics and is quite user-friendly, and thus enables the user to redesign as many or as few of the characters as desired and store the resulting font on tape or, with a two-line modification, to disc. This takes 3 sectors and twenty seconds to store on disc, and takes five minutes to save on tape (according to the instructions which come with the tape).

These are the lines to modify for disc operations:

```
1520 OPEN "1:"+F$ FOR INPUT AS #1
1770 OPEN "1:"+F$ FOR OUTPUT AS #1
```

If you want to change only a small number of characters, you may find it more convenient to write a short procedure to poke the character codes into the video RAM directly. For this you need to know that it takes eight bytes for each character and that characters 32 to 127 are stored consecutively in the Video RAM, starting at address 2048, and that characters 160 (LEFT GRPH A) to 215 are stored starting at address 3584.

When you have finished using the special font, the default font can be reinstated simply by using the command: screen 0. Consequently ensure that while you do want your special font, that any program you use does not have that statement in it - it is usually unnecessary in a text program anyway since screen 0 is the default screen.

Unfortunately Boal's program has one serious and quite unnecessary weakness: it does not allow the left or right graphic characters to be redesigned. This considerably reduces the value of the program, as it would be useful, for example, to change the graphics characters to be the same as those of whichever printer one is using.

The flaw occurs because the character fonts are stored in the video RAM in a different order from the ASCII codes. The video RAM order can be shown by Garth Thornton's one-line program (Newsletter No 3, Aug 1984, p7), although you should make allowance for the cursor which ends up in the way. You will find that the ordinary characters (ASCII 32 to 127) are followed by their reverse video versions and then by the graphics characters for which there are no reverse video versions.

If you try to redefine a left or right graphic character using Boal's program you will find it displays one of the inverse video characters instead of the one you wanted.

The following modifications cure that shortcoming completely, and I wonder why Boal did not make such a provision in his version of the program. Firstly remove the semicolon at the end of statement 2150, then key in the following lines, holding down the LEFT GRPH key for the string characters in lines 2152-4 and RIGHT GRPH in lines 2156-8:

MICROSTYLE COMPUTERS LIMITED
198 JACKSON STREET, P.O. BOX 38-479, PETONE. TELEPHONE 686-963

HOURS	9.00AM - 5.00PM	MONDAY TO THURSDAY
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	9.00AM - 1.00PM	SATURDAYS

IN ADDITION TO THE FULL RANGE OF SVI SPECTRAVIDEO PRODUCTS WE ALSO CARRY THE FULL RANGE OF "PROGRAMZ" SOFTWARE AND PRODUCTS, AS WELL AS THE FULL RANGE OF BUSINESS PROGRAMS FOR THE SPECTRAVIDEO FROM RGW SYSTEMS, THE STOCK CONTROL/ACCOUNTS PROGRAM FROM MICROSYSTEMS AND THE CARPE SERIES OF STOCK CONTROL/INVOICING, DEBTOR'S, CREDITORS AND GENERAL LEDGER FROM CARPE OFFICE SYSTEMS (N.Z.) LTD.

PROGRAMZ SOFTWARE TITLES. #18 ON TAPE #23 ON DISK

- P101 QUICKDRAW:-A SCREEN DRAWING PROGRAM SUITS CHILDREN 2 YEARS AND OVER.**
- P103 KIWI COUNTRY - NORTH ISLAND :-A WELL WRITTEN PROGRAM WHICH SHOULD BE A MUST IF YOU HAVE SCHOOL AGED CHILDREN**
- P106 TYPEMASTER:- EDUCATIONAL PROGRAM AND TYPING TUTOR**
- P107 WORLD ATLAS:-GENERAL KNOWLEDGE, EDUCATIONAL PROGRAM WHICH INCLUDES A GRAPHIC WORLD MAP(TAPE VERSION #15)**
- P108 WORLD ATLAS EX (REQUIRES EXPANDED 318):-EXPANDED VERSION OF WORLD ATLAS(P107) MORE COUNTRIES, LEADERS, CLUES AND CAPITALS AS WELL AS MOVEMENT OF SUN TIME, MONTH, YEAR SEASONS.**
- P121 SPECTRONIC CALCULATOR:-THIS VERSATILE PROGRAM INCLUDES A METRIC CONVERSION MODE AND A LISTED CALCULATION MODE AS WELL AS THE NORMAL FEATURES OF A CALCULATOR**
- P122 SKILLMAN:-EDUCATIONAL GAME PROGRAM**
- P123 BATS ALIVE:-GAME PROGRAM (TAPE VERSION #15)**
- P124 T.V. NIGHTMARE:-GAME PROGRAM "LONG TERM PLAYABILITY"**
- P141 MATHS 1 :- BEGINERS MATHS WITH SIMPLE GRAPHICS**
- P142 MATHS 2 :- CARRIES ON FROM MATHS ONE**
- P151 FLIGHT SIMULATOR(REQUIRES EXPANDED 318):- PILOT A 747 JUMBO JET AROUND NEW ZEALAND AND AUSTRALIA (AVAILABLE SOON)**

R.G.W SYSTEMS BUSINESS SOFTWARE(DISK DRIVE REQUIRED).

FINANCIAL REPORTING SYSTEM(SINGLE DISK VERSION)	\$139.00
FINANCIAL REPORTING SYSTEM(TWIN DISK VERSION)	\$155.00
JOB COSTING SYSTEM	\$65.00
STOCK CONTROL SYSTEM(80 COL CARD REQUIRED)	\$80.00

-----<< SPECTRAVIDEO >>-----
 -----<< SV EXTENDED BASIC >>-----
 -----<< ROM ROUTINES >>-----

compiled by Deane Whitmore.

The following listings are ROM entry points for the various jumps within the ROM code. Similarities between SPECTRAVIDEO BASIC and MSX/BASIC have been indicated with a "#". These particular routines have a small description in the Microsoft MSX/BASIC Handbook.

----- Rom Routines/ Entry Points -----

OUTOC 0018	FORSIC 0019	SIGN 002B	GETYPR 0030	DRMLEN 004C
FILLIN 0109	LOGOR 01C1	SINFIK 0245	COBFIK 0248	TANFIK 024D
ATNFIK 024F	ALPTAB 0295	DNVMSB 062C	DIVMSB 0684	MULTNPM 0800
OTXTCB 0800	CONSTR 084C	INTXT 069A	REDDY 06FF	BRKTXI 080A
FNDFOR 08AA	READYR 08C9	SNERR 08ED	DVDEPR 08F0	NERR 08F3
DDERR 08F6	DVERR 08FF	MDERR 0902	THERR 0905	ERROR 0907
ERESET 092E	ERRFIN 098D	STPROY 09AE	READY 09AF	REPINI 09C1
MAIN 09CA	INLIN 09FB	ESENT 0A2B	FINI 0AC9	LINKER 0AEB
CHAD 0AE9	FNOLIN 0B27	CRUNCH 0B44	RENCRN 0BF0	LDERR 0B27
NOTFRN 0B2F	NKTCOIN 0E3A	NEMSTT 0E3E	NMSTRT 0E51	ONE 0E82
BCRBTB 0EAD	CRBT2 0EAE	CHRCOIN 0EB2	INTIDK 0F99	INTID2 0F9A
PCERR 0F9E	LINESC 0FA3	LINEET 0FAD	GOSUB 0FF6	GOSUB2 1013
GOTC2 102B	USERR 103C	LETCOIN 108B	NTONB1 1186	LINPT3 12CA
LINPT4 131E	FINPRT 1365	DOASIG 1457	FRMEL 14C5	FRMFRN 14CB
FRMEL 14CA	FRCHK 14CB	TSTOP 14D9	EVAL 162D	PARCHK 16E9
ISVAR 16FD	RETVAR 1700	MAKPL 170B	MAKPS 170C	CNSET 1715
DOCTON 171A	MINPLS 17A9	BERPMN 1800	BIVDL 182E	SNFLT 183C
BIVINT 183E	DOOMF 19E7	GETINT 1A98	GETIN2 1A99	INTFR2 1A9C
BTBYC 1A45	GETBYT 1AA6	CONINT 1AA9	L1ST 1AB8	SPRATR 1B00
LIEPRT 1B05	BUFLIN 1B0E	DEL 1CBF	FRMINT 1CB9	FRQINT 1DC3
SOCPTR 1D7B	DEPTR 1E00	.C1 1E13	CSAVE 1E15	CSBSAV 1E3E
CSAVE 1E43	CLDAD 1E7F	CLDAD 1EAA	SRDAS 1F34	CASCPW 1FA7
CASBNW 1FC6	CASBNR 1FE4	BRPCL 2000	.C4B 2014	CASIN 2016
CASOUT 2026	CSROIN 203A	OFFDIO 204A	DIERR 204D	CHRTON 2059
CTMOFF 206C	CTOFF 207C	DATAN 20E3	.C4S 21BA	MDLNB 21C9
MLSDN 21F6	FETCH2 223C	FETCHR 2242	DEDFET 225F	VALSDN 226D
VALSC2 2279	VARRET 22CA	MLXED 22CC	.C2B 22D0	NEBD 22DF
SCAN1 22DF	SCAND 22EE	PRESET 232B	PSET 232D	POINT 2346
ATRSN 2390	IDELT 23B1	NEBL 23BB	YDELT 23C3	ICBY 23CE
ICBY 23D8	BLINE 23E7	DOGRPH 247C	DOGRP2 248B	.C23 24F0
HLFDE 24F4	PAINT 24FC	NEGDE 264C	CIRCLE 2652	GPUB 26AB
DRW 250A	.C24 2B41	IFUTR 2B45	GETR 2B60	INTD 2B8D
BCXQ 2B9E	NMD 2BAE	MLTQ 2BCC	.C9 2BE1	MDTOR 2BES
SOUND 2BFD	PLAY 2C24	MLEND 2CDB	GETVCP 2D46	INVTCT 2D4D
PLYTAB 2DB7	PUT 2FB1	GET 2FB4	LOCATE 2FD1	NM 3036

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.C13	6400	OUTCON	6407	OUTOLP	6415	LPTCHR	643A	FINLPT	643D
PRINTW	6446	TTYCHR	6455	LINPT1	6456	CRDCH	6463	FININL	646A
CRD	6474	CRFIN	647C	CRFIN	647D	#18DNC	6495	CKSTTP	640F
INKEY	64F3	OUTCH	6513	.C10	6510	BLTU	6520	BLTUC	6523
GETBTK	652E	REASON	6537	CHERR	6545	CHERR	6550	SCRATH	6556
SCOTCH	6557	RUNC	656A	CLEARC	6571	CLEAR	6577	STKINI	6585
STKERR	65C0	STMPRT	65E7	ONTRP	65EB	OFFTRP	65FB	STPTRP	6601
RSTTRP	660E	REDTRP	6610	SETTRP	6633	FRETRP	663D	INITRP	6653
ROTTRP	666E	RESTOR	66AE	RESFIN	66C3	STOPP	66C8	STOP	66C8
STOPFB	66DC	ENDST	66CF	CONSTP	66D9	STPEND	66E3	ENDCON	66E6
CTRPPT	6709	CTRLPT	6700	CONT	6710	TON	672F	TOFF	6730
SWAP	6735	ERRGE	676E	POPHIT	6790	ISLET	679E	ISLET2	679F
CLEAR	67A6	SUBDE	681A	NEXT	6821	#19FLD	68C2	.C15	68C7
STRDTP	68CD	STRD	68FA	STRH	68FF	STRB	6904	STR	6909
STRCPY	6916	STRINI	692A	STRINI	692C	STRAD1	6932	PUTDE1	6934
STRLIT	693A	STRLT1	6930	STRLT3	693D	STRLT2	693E	PUTNEW	6959
PUTTRP	695D	STROU1	697C	STROUT	697D	STMPRT	6980	GETSPA	6993
PPSWRT	699C	SARBA2	6900	CAT	6A0C	FRESTR	6A05	FREFAC	6A08
FRET02	6A0B	FRETRP	6A0C	FRETHS	6AF3	LEN	6804	ASC	6810
ASC2	6814	CR	6820	GETSTR	6826	FINBOX	682A	STRNG	682E
SPACE1	684D	LEFT	6866	LEFTUS	686D	RIGHT	6896	NID	689F
VAL	68D0	INSTR	68F0	LHEND	6C73	FRE	6CF7	.C12	6D11
BPINLIN	6D13	#QINLIN	6D26	#INLIN	6D2F	.C33	6FD1	WNSCON	6FD3
WNSC1	6FD6	GBNBLK	702F	GETFLP	7033	GETPTR	7036	FILCON	7067
SETFIL	7073	OPEN	7080	NULOPN	70C3	CLSFIL	70EA	NOLSB	710A
LRIN	711F	LOAD	7121	MERGE	7122	SAVE	7167	SFSVEX	71AB
CHKTOP	7209	NORCON	7210	GETDEV	721A	RSET	7227	LSET	722B
FIELD	722D	HK1	7312	MS	7315	MD	731B	CVI	732B
CVS	732E	CVD	7331	CLOSE	7375	CLBALL	737D	CLBLR	7380
LFILES	73AD	FILES	73B2	OPUT	7380	DEET	7389	FILEUT	73C9
FILEU1	73CA	INDSKC	73F1	INDSKE	7402	FIXINP	7406	CLBLF	7469
DOCLR	7474	GETBLF	747A	GETBF1	747D	LOC	748A	LIF	749A
EOF	7480	FPOS	74C6	DROB	74D9	FILINP	74E6	FILEBT	74EB
PRGFIN	750C	FILIND	7514	DLNE	7520	NDKCR	75D0	DERBFM	75FA
DERFAD	75FD	DERFOR	7600	DERFV	7603	DERFND	7606	DERFV	7609
DERIFN	760C	DERIER	760F	DERPE	7612	DERGAP	7615	DERSD	761B
BSAVE	7624	BLOAD	7684	CHKBRN	76EA	.C6	7706	PARDEV	770B
DEVTL	7780	GENOSP	77AB	.C3	77BA	KBDOSP	77CC	CRTOEP	7817
CASDSP	7841	MDMDSP	791F	LPTDSP	7953	.C29	798B	CHKMD	798E
DIAL	79C2	SMTRP	79CB	RCVX	79D3	RS2INT	79D8	.C2B	79D9
BOOT	79DC	.C47	7A53	INITAT	7A66	FWKRON	7A84	.C30	7B3E
MDN	7B44	MDERR	7B4A	.C27	7B4E	INIT	7B50	INIENT	7B9F
SETMAX	7CBA	DEFILE	7CDA	.C5	7D44	LASTMR	7D80	C000	C000

--- BIOS WORK AREA LIST

INITSA	F500	D3	RANLW	F500	D3	RNDCH	F504	00	RNDTAB	F506	35
#ERTAB	F52B	9E	#ERRFLB	F53F	02	LPTLST	F540	00	#LPTPOS	F541	00
#PRTFLB	F542	00	BLINLEN	F543	50	RUBSN	F545	00	ISTKTOP	F546	F0
#JURLIN	F548	32	#TXTTAB	F54A	01	#ALZADR	F54C	00	#KBLF	F54F	20
#BUFIN	F680	C3	#BLF	F68E	20	#ENDBLF	F790	00	#TTYPOS	F791	00
#DIPLFB	F792	00	#VALTYP	F793	02	#CONSFW	F798	00	#WEGIZ	F7A2	A0
#TEMPPT	F7A4	A6	#TEMPST	F7A6	06	#DISCTP	F7C4	06	#DISCPTR	F7C5	67
#FRETCP	F7C7	68	#TEMP3	F7C9	36	#TEMPB	F7CB	36	#ENDFOR	F7CD	0F
#SUBFLB	F7D1	00	#USFLB	F7D2	FF	#TEMP	F7D3	00	#PTRFLB	F7D5	00

GAUTFLB	F7D6	00	QSAVITX	F7D8	12	QSAVSTK	F7D0	EE	WBRFLN	F7DF	14
WDOT	F7E1	1E	WONELIN	F7E3	00	WONEFLB	F7E7	00	QTEHP2	F7EB	36
WOLDLIN	F7EA	0A	WOLDTXX	F7EC	00	QWARTAB	F7EE	44	WARYTAB	F7F0	4A
WSTREND	F7F2	4A	WDAIPTR	F7F4	00	WDEFTBL	F7F6	08	WPRVSTK	F810	00
WPRULEN	F812	00	WPARMI	F814	4C	WPRMPRV	F818	10	WPRMLJ2	F87A	00
WPRVFLB	F8E0	00	WARYTAZ	F8E1	4A	WDFUNB	F8E3	00	QTEHP9	F8EA	10
WFUNACT	F8E6	00	WULZDAT	F8E8	46	WQAPTHP	F8E9	22	WTRDFLB	F8F1	00
WFLUFR	F8F2	2B	WFLTT1	F91A	33	WFLTT2	F91B	34	WDECTHP	F91D	26
WDECTM2	F91F	73	WDECCNT	F921	01	WDAC	F923	42	FAC	F923	42
WACLD	F925	25	WHLDB	F933	10	WLDL5	F948	00	WHL02	F963	00
WHL0	F968	00	WAB8	F974	C3	WANDX	F984	00	WAXDRV	F98C	02
WAVFIL	F98D	01	WFLTAB	F98E	A2	WRTAB	F990	FA	WMLBUF	F992	AF
WLDREV	F994	00	WDRVTR	F995	00	WTRFIL	F997	00	WREPLC	F999	AF
WSTFRE	F99B	10	WFLMOD	F99D	00	WMLFLB	F99D	00	WFLNWH	F99E	6C
WFLNWH	F9A7	E0	WSTTRK	F9B0	E9	WSTBCT	F9B1	EA	WNLNLY	F9B2	00
WAVFLB	F9B3	EC	WBAVEND	F9B4	ED	WDBSY	F9B6	EF	WBRONT	F9B7	00
WBRONT	F9B8	00	WAVFLB	F9B9	00	WDBFLB	F9BA	F3	WAVEBC	F9BB	F4
WSTAT0	F9BC	F3	WSTAT1	F9BD	F6	WSTACK	F9BE	F7	WCRST0	F9BD	4A
WTRAM	FA00	4A	WOLIKSW	FA02	01	WCSRY	FA03	17	WCSRX	FA04	3E
WCSRW	FA05	01	WCSDFB	FA06	FF	WBSRW	FA07	F0	WTRFLB	FA08	30
WDFL0	FA09	01	WDFCLR	FA0A	0F	WDFCLR	FA0B	04	WDFCLR	FA0C	07
WAVLUPD	FA0D	C3	WMLNUPD	FA10	C3	WATRYBT	FA13	0E	WUTFN	FA14	C3
WQUEUES	FA17	F2	WREPCNT	FA19	0C	WPUTPNT	FA1A	96	WGETPNT	FA1C	96
WFWSTR	FA1E	66	WDFL0	FA20	20	WCMERK	FA2F	63	WCHROM	FA2D	2A
WLOC	FA2D	2A	WCMERK	FA2C	00	WASPECT	FA23	D3	WCCNONT	FA25	DB
WOLINEF	FA27	4F	WCMFNTB	FA2B	E6	WDFLOT	FA2A	D3	WDFONT	FA2B	EC
WDFONTB	FA2D	00	WDCSUN	FA2F	7D	WDCSTONT	FA21	F3	WDCSLY	FA23	06
WCSAVEA	FA24	7C	WCSAVEN	FA26	31	WCRDFF	FA27	CA	WCRDFF	FA29	00
WLOHNEK	FA28	D3	WLOHDIR	FA2C	8C	WLOHDIR	FA2D	C9	WLOHONT	FA2F	00
WSPONT	FA2E	00	WDMONT	FA25	00	WDFOTREC	FA25	00	WDFPROB	FA26	00
WRTTPOB	FA27	00	WUTFLB	FA2B	00	WMLNDEL	FA29	00	WMAXDEL	FA2B	00
WRTPTR	FA2D	00	WMLTAB	FA2F	00	WMLFLB	FA21	00	WMLTAB	FA22	00
WQUEBK	FA2A	00	WVOICD	FA2E	00	WVOICD	FA2E	00	WVOICD	FA2F	00
WRS21D	FA2E	00	WPRDONT	FA2E	00	WBAWSP	FA2F	00	WVOICEN	FA21	00
WBAWOL	FA2D	00	WMLLEN	FA2D	00	WMLPTR	FA2D	00	WQUELEN	FA2D	00
WMLSTCF	FA2B	00	WPLYONT	FA29	00	WVCR	FA2A	00	WVCR	FA2F	00
WVCR	FA24	00	WMLFLB	FA29	00	WSTPOY	FA2A	00	WFRONW	FA2B	00
POLTRY	FA2C	00	WMLNTR	FA2D	AF	WSTPOB	FA25	68	WMLDSAV	FA27	00
WFKSWI	FA28	01	WFKFLB	FA29	00	WMLNGBF	FA23	00	WMLIKFL	FA24	00
WMLKEY	FA25	FF	WMLNKEY	FA2B	FF	WMLTKEY	FA26	FF	WMLYBLF	FA2B	6C
WMLFEND	FA2B	00	WMLNARK	FA24	FF	WMLTARK	FA2C	00	WMLBTTON	FA2A	00
WMLNEN	FA26	BB	WMLTSAV	FA2B	00	WMLTSAV	FA2A	00	WMLTPTBL	FA2B	00
WMLTONT	FA2A	00	WMLNTR	FA2B	00	WMLT	FA2C	00	WMLT	FA2D	00
WMLTFF	FA2E	EA	WMLNTR	FA2B	00	WMLNTR	FA22	13	WMLNTR	FA2A	00
WMLFLB	FA25	00	WMLNTR	FA26	00	WMLSTYLE	FA27	00	WMLPST	FA2B	00
WMLNEN	FA29	00	WMLNTR	FA2A	00	WMLTIZ	FA2B	00	WMLSAV	FA2C	00
WMLSTATFL	FA2D	C7	WMLNTR	FA2E	00	WMLNTR	FA2F	00	WMLNTR	FA2D	00
WMLNTR	FA21	00	WMLNTR	FA22	00	WMLNTR	FA24	00	WMLNTR	FA26	00
WMLNTR	FA24	00	WMLNTR	FA2A	00	WMLNTR	FA2B	00	WMLNTR	FA2C	00
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WMLNTR	FA2D	00	WMLNTR	FA2E	00	WMLNTR	FA2F	00	WMLNTR	FA2D	00
WMLNTR	FA2E	00	WMLNTR	FA2F	00	WMLNTR	FA2D	00	WMLNTR	FA2E	00
WMLNTR	FA2F	00	W								

--- Hook Jump Table

HL.KEY1	FE79	HL.PRTF	FE7C	HL.DGET	FE7F	HL.INDS	FE82	HL.BONE	FE85
HL.GNDF	FE8B	HL.FFOS	FE8B	HL.READ	FE8E	HL.ISRE	FE91	HL.MAIN	FE94
HL.RSLF	FE97	HL.LOC	FE9A	HL.BAKU	FE9D	HL.BTKE	FEA0	HL.PARD	FEA3
HL.FRET	FEA6	HL.NTFL	FEA9	HL.NTFM	FEAC	HL.CLEA	FEAF	HL.GAND	FEB2
HL.SAVE	FEB5	HL.FILE	FEB8	HL.LOF	FEBB	HL.NTFL	FEBE	HL.MODE	FEBC
HL.DGR	FECA	HL.MERG	FECD	HL.ELF	FECA	HL.PTRG	FECD	HL.NFO	FEDE
HL.PRGE	FECD	HL.BUFL	FEDE	HL.CROD	FEDE	HL.CKND	FEDE	HL.GETP	FEDE
HL.LOPD	FEED	HL.DEWN	FEED	HL.SETF	FEED	HL.NLO	FEED	HL.RETU	FEED
H.CJRC	FEF1	HL.LIST	FEF4	HL.RUNC	FEF7	HL.EVAL	FEFA	HL.ISNI	FEFD
HL.COMP	FF00	HL.FRQI	FF03	HL.DIRD	FF06	HL.CUTD	FF09	HL.NDTR	FF0C
HL.GEND	FF0F	HL.FILO	FF12	HL.ISFL	FF15	HL.ERRP	FF18	HL.ENRF	FF1B
HL.TRN	FF1E	HL.CRUS	FF21	HL.CRUN	FF24	HL.FINP	FF27	HL.FRME	FF2A
HL.BINS	FF2D	HL.FINI	FF30	HL.BINL	FF33	HL.FINE	FF36	HL.FINB	FF39
H.INDI	FF3C	HL.WIDT	FF3F	HL.PINL	FF42	HL.GINL	FF45	HL.INLI	FF48
HL.DSKC	FF4B	HL.ERAF	FF4E	HL.DSPF	FF51	HL.NEWS	FF54	HL.GONE	FF57
H.DMT	FF5A	H.MMD	FF5D	H.MDC	FF60	H.MDM	FF63	H.MMD	FF66
H.MNE	FF69	H.MMD	FF6C	H.DIAL	FF6F	H.RS21	FF72	HL.ONSD	FF75
HL.KYCL	FF7B	HL.KYEA	FF7B	HL.NMI	FF7E	HL.KEYC	FF81	H.MDM	FF84
H.BADC	FF87	HL.DSKD	FF8A	HL.SETS	FF8D	HL.NME	FF90	HL.KILL	FF93
HL.IPL	FF96	HL.COPY	FF99	HL.CMD	FF9C	HL.DGXF	FF9F	HL.DSKI	FFA2
HL.ATR	FFA5	H.MNE	FFA8	H.INIP	FFAB	HL.CPU	FFAE	HL.TOTE	FFB1
ENDMK	FFB4								

Ok all you machine code freaks , lets start disassembling this lot and letting us what some of these routines do , what registers need to to set with what prior to calling routines and what they return if anything. Thanks for the list Deane ! .



KAPITI MEETING - RAUMATI SONJTA MEMORIAL HALL, TENNIS COURT RD, RAUMATI

FROM PAGE 18

```
800 IF ASC(A#)<160 THEN  IA=2048+8*(ASC(A#)-32) ELSE IA=3584+8*(ASC(A#)-160)
805 FOR I=IA TO IA+7
1230 IF ASC(E#)<160 THEN  IA=2048+8*(ASC(E#)-32) ELSE IA=3584+8*(ASC(E#)-160)
1235 FOR I=IA TO IA+7
2152 PRINT: PRINT"a b c d e f g h i j k l m n o p q r s ":'left graphics
2154 PRINT: PRINT"t u v w x y z ":'left graphics
2156 PRINT: PRINT"a b c d e f g h i j k l m n o p q r s ":'right graphics
2158 PRINT: PRINT"t u v w x y z ":'right graphics
2370 IF ASC(E#)<160 THEN  IA=2048+8*(ASC(E#)-32) ELSE IA=3584+8*(ASC(E#)-160)
2375 FOR I=IA TO IA+7
```

FROM PAGE 17

The next broadcasts for SPECTRAVIDEO are: 6th April, 4th May, 1st June in other words the 1st Saturday in the month at 9am on station ZYB (783 kHz). Mark the calendar.

I'd be interested to hear from any of you receiving the broadcasts outside the Wellington area. Anyone going to try from the Chatham Islands!

Garry Clark
Access Radio, Box 2396
Wellington

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WGTM SV USERS GROUP

Bug Report Form

Please fill in the following and send to BUGS , P.O Box 7057 ,
Wgtn South , Wellington

Only 1 bug per form please

1) Where bug occurs

a) Basic Command --> Which Command -----

b) Commercial Basic Program --> Program Name -----

Line Number if applicable -----

c) Commercial Machine Code Program --> Program Name -----

d) CP/M Program --> Program Name -----

e) Hardware --> Hardware Name and Number -----

f) Manual / Book --> Title -----

Author ----- Page -----

g) Other .. Please specify

2) Describe the bug , and under what circumstances it occurs if possible

+++++

3) If You have a fix for the bug please describe it here or on a separate sheet

Serial Number Safekeeping

Membership Number _____

Enter your serial numbers beside the relevant hardware number

SV318 _____

SV328 _____

SV903 Cassette _____ SV904 Casse _____

SV806 80 Col _____ SV805 RS232 _____

SV803 16k Ex _____ SV807 64k _____

SV605 Expander _____ Is expander 605/605A/605B __

SV601 Expander _____ SV902 Drive _____

SV801 Controll _____ SV802 Centr _____

Printer Type _____ Serial Numb _____

Monitor Type _____ Serial Numb _____

Do not enter name or address on this form .

This form will be kept on separate premises to the Club Member list.
